

CoRoT-2b

R-0074

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_250504 · created 2026-07-15T21:33:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460799.9156 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.222 +/- 0.066
Transit depth (Rp/Rs)^2	4.94 +/- 2.92 [%]
Orbital Inclination (inc)	89.11 +/- 2.03
Airmass coefficient 1 (a1)	1.12 +/- 0.5
Airmass coefficient 2 (a2)	-0.22 +/- 0.2
Scatter in the residuals of the lightcurve fit is	41.97 %
Best Comparison Star	None
Optimal Aperture	5.21
Optimal Annulus	5.21
Transit Duration (day)	0.0995 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.222 ± 0.066 vs 0.1667 (deviation 0.84σ)
Duration fit vs published	0.0995 d vs 0.095 d (deviation 0.67σ)
Mid-time O–C	-6.6 min
Depth SNR	3.4
Residual scatter	41.97 %

Light curve

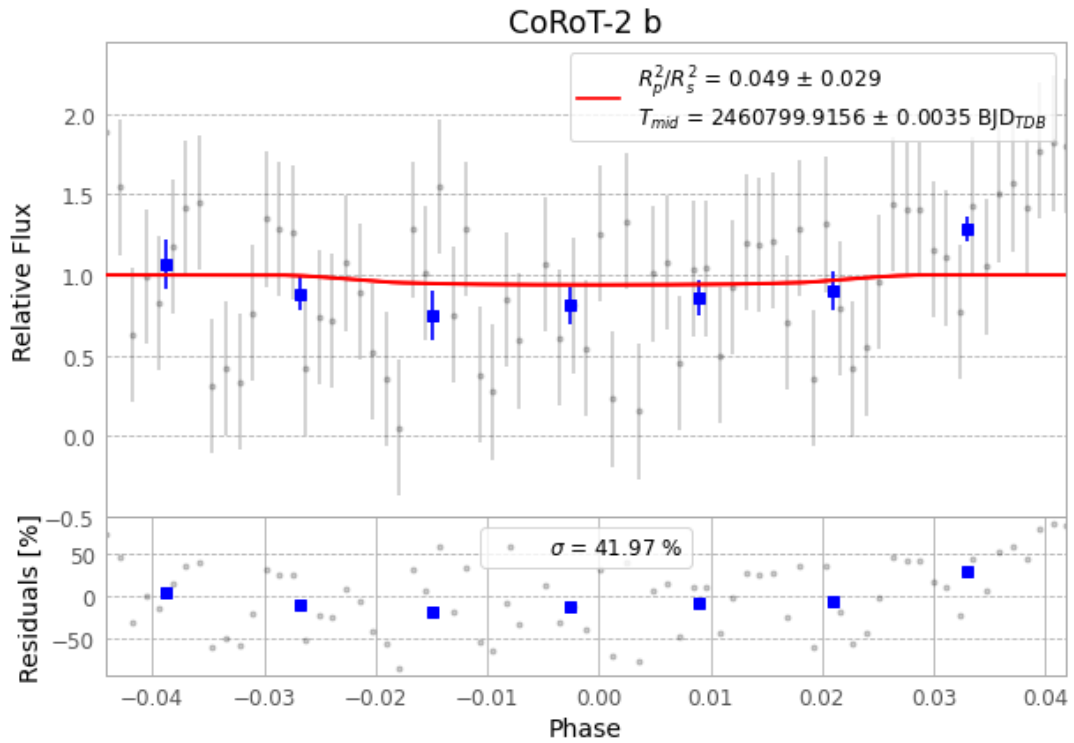


Figure 1 — FinalLightCurve_CoRoT-2 b_2025-05-04.png (SHA-256 bdd313def6ba0be2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [36, 424], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f30f092a52efc481...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

73 FITS frames (48 MB), CoRoT-2250504080315.FITS → CoRoT-2250504113915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-250504.log (c710ca66f6ba...), FinalLightCurve_CoRoT-2 b_2025-05-04.png (bdd313def6ba...), FinalLightCurve_CoRoT-2 b_2025-05-04.csv (317c95509bfe...)

Compute resources

Wall time 155.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:33Z.